Ex.No. 2: Implementing TDMA Slot Allocation in GSM

AIM: To implement TDMA (Time Division Multiple Access) slot allocation in GSM using

MATLAB and analyze system performance in terms of time slot duration, frame utilization,

and number of allocated slots.

Objective:

1. To calculate the Time Slot Duration in a GSM TDMA frame by dividing the total frame

time into equal time slots. This helps in understanding the time structure of GSM

communication.

2. To determine the Frame Utilization (%) based on the number of allocated time slots

compared to total available slots. This evaluates the efficiency of TDMA resource usage.

3. To compute the Number of Allocated Time Slots for multiple users in the GSM system.

This ensures proper scheduling and interference-free communication.

Procedure:

1. Define GSM frame parameters such as total frame duration and number of time slots.

2. Compute time slot duration by dividing frame duration by number of slots.

3. Assign time slots to multiple users based on availability.

4. Calculate frame utilization percentage using allocated slots vs total slots.

5. Determine total number of allocated slots for active users.

6. Plot time slot allocation and utilization using MATLAB graphs.

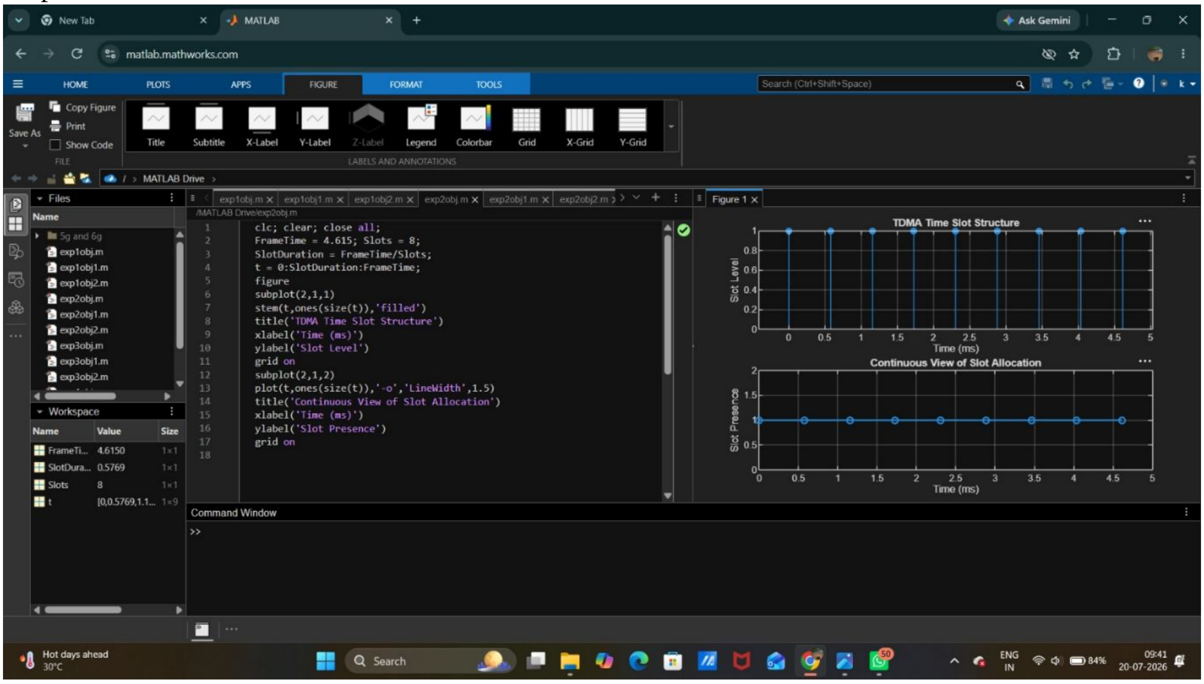
7. Display results in the command window for analysis..

Objective1:

Program:

```
clc; clear; close all;
FrameTime = 4.615; Slots = 8;
SlotDuration = FrameTime/Slots;
t = 0:SlotDuration:FrameTime;
figure
subplot(2,1,1)
stem(t,ones(size(t)),'filled')
title('TDMA Time Slot Structure')
xlabel('Time (ms)')
ylabel('Slot Level')
grid on
subplot(2,1,2)
plot(t,ones(size(t)),'-o','LineWidth',1.5)
title('Continuous View of Slot Allocation')
xlabel('Time (ms)')
ylabel('Slot Presence')
grid on
```

output:



Objective2:

Program:

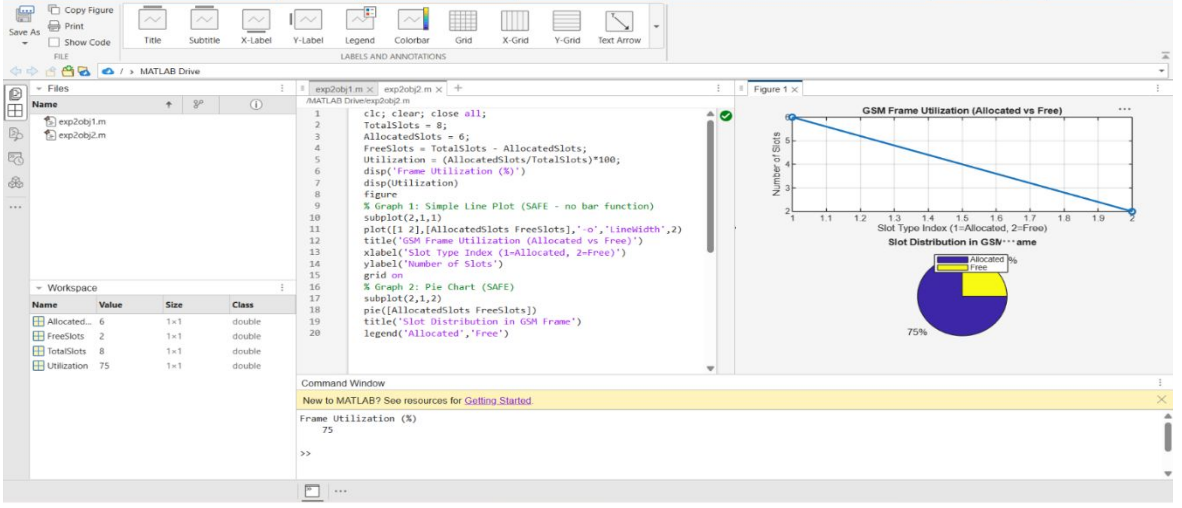
```
clc; clear; close all;

TotalSlots = 8;

AllocatedSlots = 6;

FreeSlots = TotalSlots - AllocatedSlots;

Utilization = (AllocatedSlots/TotalSlots)*100;

disp('Frame Utilization (%)')

disp(Utilization)

figure

% Graph 1: Simple Line Plot (SAFE - no bar function)

subplot(2,1,1)

plot([1 2],[AllocatedSlots FreeSlots],'-o','LineWidth',2)

title('GSM Frame Utilization (Allocated vs Free)')

xlabel('Slot Type Index (1=Allocated, 2=Free)')

ylabel('Number of Slots')

grid on

% Graph 2: Pie Chart (SAFE)

subplot(2,1,2)
```

pie([AllocatedSlots FreeSlots])

title('Slot Distribution in GSM Frame')

legend('Allocated','Free')

Output:



Objective3:

Program:

clc; clear; close all;

Users = 1:8; % Total users

TotalSlots = 5; % Available slots

Alloc = zeros(1,8); % Initialize allocation

Alloc(1:TotalSlots) = 1; % Assign slots to first users

Blocked = 1 - Alloc; % Remaining users blocked

disp('User Wise Slot Allocation')

disp(table(Users',Alloc','VariableNames',{'User','Status(1=Allocated,0=Blocked)'}))

figure

% Graph 1: Allocation per user

…% Graph 2: System view (safe alternative to bar)

subplot(2,1,2)

plot(Users,Alloc,'-o','LineWidth',2)

hold on

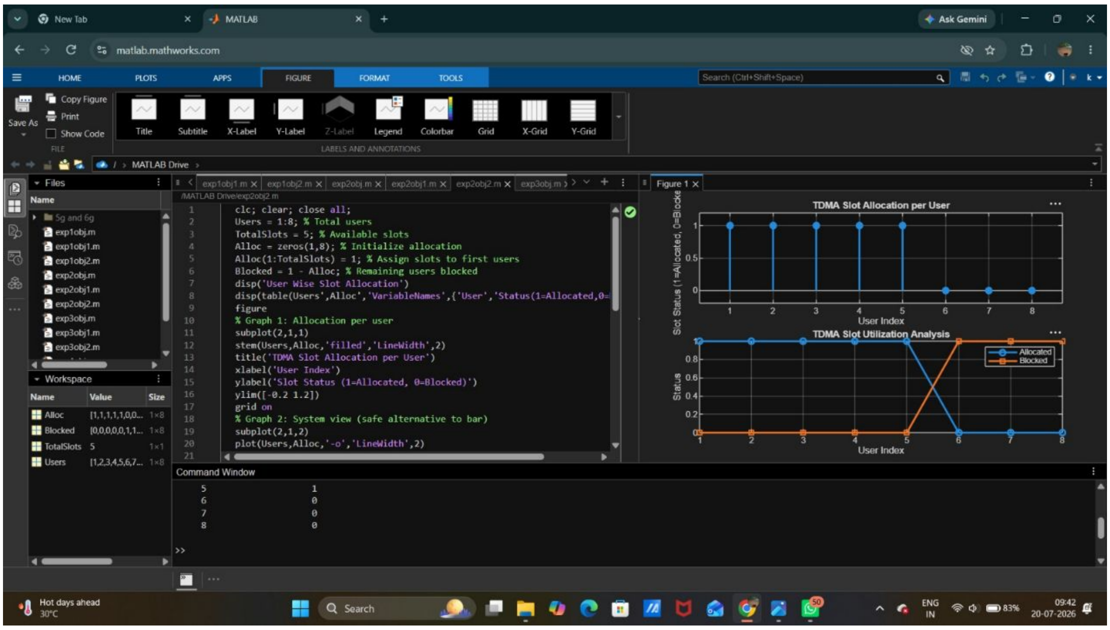
plot(Users,Blocked,'-s','LineWidth',2)

title('TDMA Slot Utilization Analysis')

xlabel('User Index')

```
ylabel('Status')
legend('Allocated','Blocked')
grid on
```

Output:



Result:

1. The time slot duration in the GSM TDMA frame was successfully calculated, confirming equal division of the frame into fixed time intervals.

2. The frame utilization was effectively analyzed, showing the proportion of allocated and free slots in the TDMA system.

3. The slot allocation per user was successfully implemented, demonstrating that limited slots are assigned to users while excess users are blocked due to resource constraints.